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# CIVIC COMPLAINTS REGISTRATION APPLICATION FOR CITIZENS OF RURAL/URBAN AREAS

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## ABSTRACT

Citizen engagement and technology usage are two emerging trends driven by smart city initiatives. Typically, citizens report issues, such as broken roads, garbage dumps, etc. through web portals and mobile apps, in order for the government authorities to take appropriate actions. Several mediums – text, image, audio, video – are used to report these issues. Through a user study with 13 citizens and 3 authorities, we found that image is the most preferred medium to report civic issues. We aim Through this platform, we aim to achieve the following: Enable citizens to easily submit complaints, particularly via image-based submissions, Automate the complaint handling process, from Geo-fencing tagging to department allocation, Utilize AI, ML, and Data Science to enhance efficiency and accuracy in complaint resolution, Foster a sense of community engagement and participation in local governance. Method which implemented are Image Processing, Computer Vision and Adversarial Scene Graph Model: To capture and interpret complaint images, Geo-fencing: To precisely determine the location of each issue, AI and ML: To automatically categorize complaints and allocate them to the relevant departments, Data Science: To analyze patterns and trends, aiding in efficient resource allocation, User-Centric Design: Ensuring an intuitive and accessible interface for citizens. Citizens can report issues promptly through images, reducing reporting barriers. The Geo-fencing and automated department allocation result in faster response times. AI and ML technologies ensure accurate tagging and categorization of complaints. By harnessing the power of automation and technology, we enable citizens to actively participate in improving their local communities. This application not only streamlines the process of addressing complaints but also promotes transparency and civic engagement.

**Keywords:** Civic Issues, Image Processing, Complaint Redressing, Geo-Tagging, Geofencing.

## I. INTRODUCTION

"Civil Complaint Registering Application" designed for both urban and rural areas. By harnessing the power of Image Processing technology, we capture images of issues, automate key details such as address, pin code, and geotagging, and categorize complaints based on image analysis. This streamlined process ensures that each complaint is efficiently routed to the respective department for swift resolution, revolutionizing the way community issues are addressed.

## II. RELATED WORK

**Kumar Prasad, Ritesh Patil, Sagar Beldare, Prof. Anita Shinde Civic Complaint Application under Smart City Project Published on (IJCSIT) International Journal of Computer Science and Information Technologies, Vol. 7 (1) ,ISSN: 0975-9646 2016, 402-406**

In this paper, the paper proposes the development of an online web application, accessible through mobile devices, that allows citizens to report problems in their city to the relevant authorities. The application utilizes GPS sensors to pinpoint the exact location of the issue, and users can provide visual proof through images. The system then generates a comprehensive report that includes user-entered data, location information, and visual evidence, which is sent to a central server for notification to the concerned authority.[8]

**Shanu Kumar, Anjali Singh, Mohit Jain Adversarial Adaptation of Scene Graph Models for Understanding Civic Issues Published on (International World Wide Web Conference Committee), published under Creative Commons CC-BY 4.0 License. ACM ISBN 978-1-4503-6674-8/19/05.**

In this paper, the Adversarial graphs contain a set of objects and semantic relations between them, specifically representative of the underlying civic issue. The paper also releases two multi-modal datasets containing text and images to facilitate further research in this area. To address the lack of labeled training data for civic issues,

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the paper uses adversarial adaptation techniques to repurpose existing scene graph models for this new task. It proposes an approach to train the model using partially annotated and unpaired data.[4]

**Pengfei Xu, Xiaojun Chang, Ling Guo, Poyao Huang, Xiaojiang Chen, Alexander G. Hauptmann A Survey of Scene Graph: Generation and Application Published on IEEE Trans. Neural Netw.Learn. Syst. 2020.**

In this paper, Authors introducing scene graphs as data structures used to describe objects, their attributes, and relationships in a scene. It highlights the importance of scene graphs in various visual tasks, such as image retrieval, image/video captions, Visual Question Answering (VQA), image generation, and specific relationship detection. The paper categorizes scene graph generation methods based on the inference models for visual relationship detection and discusses the applications of scene graphs in specific visual tasks. It also highlights some of the challenges and future research directions in the field of scene graph generation.[3]

**Anjaly Antony, Minla K.S Civic Complaint Reporting Using Image Processing Published on IJCRT — Volume 9, Issue 6 June 2021 — ISSN: 2320-2882**

In this paper, the key advantages of this system include reducing the effort and time required for registering complaints, tracking the location of the issue using GPS, generating deadlines for issue resolution, and allowing citizens to vote on complaints to prioritize them. By involving citizens in the process, it encourages active participation in improving the city's infrastructure and services.[2]

**Dnyanesh Walwadkar, Saurav Yadav Smart Civic Issue Reporting System Published on International Journal of Advanced Research in Science Communication and Technology DOI:10.48175/IJARSCT-2659**

In this paper, the architecture of the system involves multiple components, including the user interface (Android app), a central server, and the involvement of citizens and officials. The app allows users to register complaints and track their status. When a complaint is registered, it is analyzed using the hybrid CNN + RNN algorithm for severity, and the SVM-NLP model processes the text description of the issue. The GPS sensor on mobile devices pinpoints the location, and the camera is used to capture visual evidence. The system then sends the data and evidence to a central server, which notifies the relevant authorities for action.[1]

## III. METHODOLOGY

The goal is to create "Civil Complaint Registering Application" we focus on a flexible communication between public people and municipal authority. Here we can also communicate with citizen to citizen and they can share their ideas and suggestions. The "CIVIC COMPLAINT REPORTING SYSTEM" advanced mobile application designed to streamline and automate the process of complaint registration and resolution. This system aims to empower citizens by allowing them to easily report issues such as garbage or damage, using their smartphones. The application utilizes image processing techniques, including the Faster R-CNN algorithm, to automatically interpret and categorize the problem, assign it a priority and severity level, and forward it to the relevant department for resolution. Additionally, the system incorporates location data obtained from the Google Maps API to provide accurate address details for the reported issue. This paper outlines the methodology adopted for the development of this application, along with a detailed flow of the complaint management process. Using this complaint system, the problems are resolved within the date and time. Firstly, people wants to complaint regarding any civic issue, then that person has to login to their account and then they can register the complaint .After that, any citizen can posts complaints regarding any civic issue, then that complaint goes to the particular department authority. That categorization of complaints by particular department authority is done by using the machine learning and image processing. Here we use the Faster R-CNN algorithm. The higher authority of that particular department can view all complaints. When a person register an issue, at the same time the system will automatically generate a date and time. By harnessing the power of Image Processing technology, we capture images of issues, automate key details such as address, pincode, and geotagging, and categorize complaints based on image analysis. This streamlined process ensures that each complaint is efficiently routed to the respective department for swift resolution, revolutionizing the way community issues are addressed.

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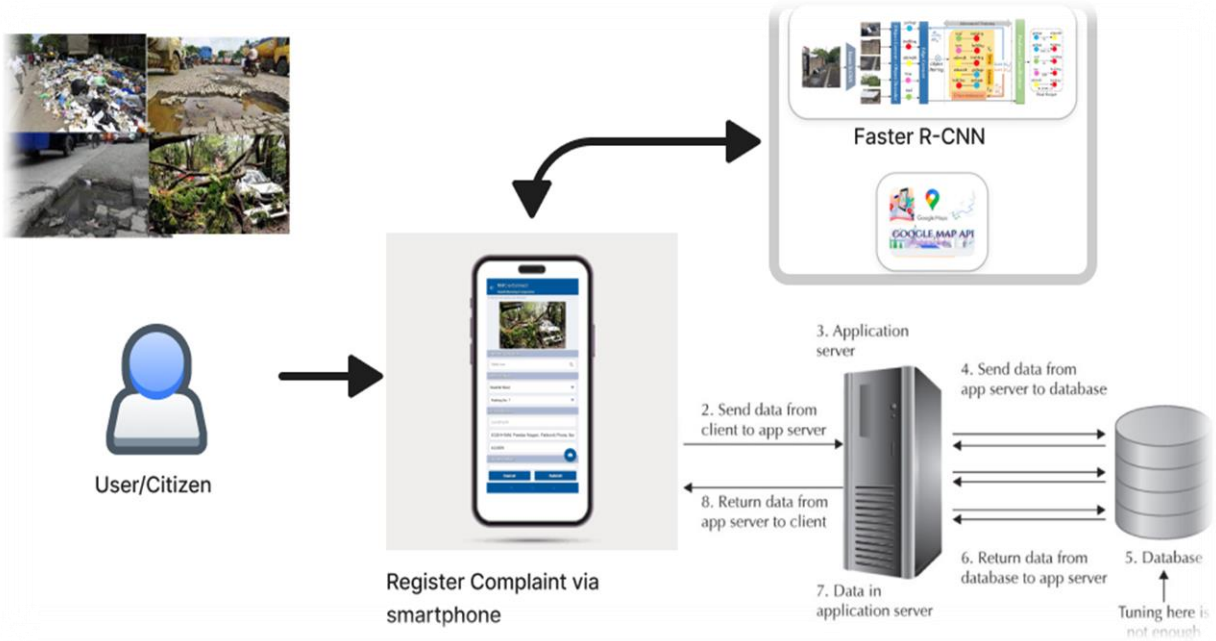



**Figure 1:** System architecture

**1)** Citizen Identification: A concerned citizen identifies an issue such as garbage or damage.

**2)** Register Complaint via Smartphone:

a) The user captures a photo of the problem and utilizes the mobile application to register a complaint.

b) The captured image is processed using the Faster R-CNN algorithm, which interprets the image, extracting data such as problem type, priority, and severity, and automatically fills in the complaint details.

c) Based on the image type, the system assigns the responsible department to handle and resolve the complaint, automatically populating the corresponding information.

d) The user's location is obtained using the Google Maps API, and the address details are added to the complaint.

**3)** Complaint Submission: The user reviews the automatically filled complaint details for accuracy. If necessary, corrections can be made by the user before submitting the complaint.

**4)** Complaint Processing: The submitted complaint is processed on the server-side, where it is assigned to the appropriate authority based on the complaint details.

**5)** Complaint Resolution: The responsible authority takes action to resolve the complaint, providing proof of resolution within the system.

**6)** Notification and Verification: The citizen receives a notification regarding the resolution status of their complaint. They have the option to verify the resolution and provide feedback and suggestions, thus closing the complaint.

## IV. ANALYSIS

In this paper, we proposed a we propose a personalized exercise and diet plan recommendation system for individuals engaging in gym activities. The analysis focuses on leveraging user-specific data, including fitness goals, current fitness levels, dietary preferences, and any health constraints, to tailor fitness regimens. By employing machine learning algorithms, the system aims to optimize dietary suggestions, ensuring a more effective and sustainable approach to achieving personal fitness goals while adapting to user's evolving needs and progress. This personalized approach not only enhances user engagement and motivation but also contributes to the overall effectiveness of fitness programs, promoting healthier lifestyles. The paper underscores the potential of such systems in revolutionizing the traditional one-size-fits-all approach to gym workouts and nutrition planning, providing a more efficient and user-centric solution in the pursuit of individual fitness objectives.

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**Advantages**

1. Enhanced Civic Engagement: Encourages citizen participation and community involvement in addressing local issues.
2. Improved Administrative Efficiency: Streamlines complaint handling and resource allocation, reducing administrative workload and costs.
3. Transparency and Accountability: Promotes transparency in local governance through real-time tracking and accountability mechanisms.
4. Cost Savings: Reduces administrative expenses by automating processes and optimizing resource allocation.
5. Community Betterment: Contributes to cleaner, safer, and more responsive environments, enhancing the quality of life.
6. User Satisfaction: Provides a convenient and responsive platform, leading to higher user satisfaction.
7. Efficient Resource Allocation: Ensures that complaints are promptly routed to the appropriate departments, optimizing government resources.
8. Reduces Response Times: Accelerates complaint resolution by automating issue categorization and resource allocation.

**Applications**

1. Local Governance Enhancement: Municipalities and local government bodies can use this application to efficiently manage and address citizen complaints related to infrastructure, waste management, and other local issues.
2. Urban and Rural Complaint Management: Streamlining and automating the handling of complaints related to waste, infrastructure, and more in both urban and rural areas.
3. Rural Development Agencies: Rural development agencies can utilize the application to facilitate citizen engagement in rural areas, focusing on agriculture, infrastructure, and basic amenities.
4. Environmental Protection Agencies: Environmental protection agencies can employ the application to address and manage complaints related to pollution, illegal waste disposal, and environmental hazards in urban and rural settings.
5. Community Organizations: Non-governmental organizations (NGOs) and community based organizations can adapt the application to encourage community members to report issues and participate in community development projects.
6. Emergency Services: The application can be used during emergencies or natural disasters to allow citizens to report urgent issues and help emergency services respond quickly to crises in both urban and rural areas

## V. OUTCOME

The system will leverage automation technology, including Machine Learning (ML), Data Science, Computer Vision and Image Processing, to streamline the process of addressing citizen complaints related to waste materials, road path holes, pipelines, drainage systems, and various other urban and rural issues. Enhanced Civic Participation: The application significantly increased citizen involvement in addressing local issues. An upsurge in registered complaints and active participation indicated a positive shift toward civic responsibility and engagement. Streamlined Administrative Processes: Administrative efficiency improved notably with streamlined complaint handling and resource allocation. Real-time tracking mechanisms reduced resolution times, optimizing administrative efforts and resources. Transparency and Accountability: The system's real-time tracking and reporting mechanisms enhanced transparency in local governance. Citizen-accessible complaint status updates and accountability measures fostered trust between administrators and citizens. Cost Savings and Resource Optimization: Automation of processes and optimized resource allocation resulted in considerable cost savings for administrative bodies. Reducing bureaucratic redundancies and response times translated into efficient resource utilization. User Satisfaction and Engagement: User feedback indicated high satisfaction with the platform's convenience and responsiveness. The user-friendly interface contributed to increased engagement and active participation in local issue resolution.

## VI. CONCLUSION

The Civic Complaint Registration Application presents a robust solution to bridge the gap between citizens and local administrations. By leveraging technology, the system not only addresses complaints effectively but also fosters a culture of civic responsibility and active community participation. The comprehensive analysis highlights the system's efficiency, user-centric design, and positive impact on local governance. However, challenges like the digital divide and data security concerns necessitate ongoing improvements and stringent measures for inclusive and secure civic engagement. Overall, the application stands as a catalyst for enhancing citizen-government collaboration and fostering sustainable community development.

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